



Commercial Auto Risk Scoring Methodology

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Business Problem

Historical data: 2020–2022 with claims outcomes

Scoring population: 2023–2024 without outcomes

Two coverage types: APD and AL

Key challenge: predict future risk using only information available at policy inception



EDA - Data Quality

~15K historical policies

Duplicate policy IDs: 75

42 exact duplicate rows identified and removed

Additional duplicate policy IDs were investigated and resolved after standardizing categorical values

Missing 88% post-claim variables, 3-8% other features

EDA - Data Quality

33 duplicate `policy_id` records were identified

Differences were caused by inconsistent capitalization in `business_type` (e.g. `trucking` vs. `Trucking`)

Standardized categories using case normalization

After standardization, duplicate records were removed

```
POL-000567 -> business_type : ['trucking', 'Trucking']
POL-001018 -> business_type : ['Other', 'other']
POL-001247 -> business_type : ['Construction', 'construction']
POL-001298 -> business_type : ['retail', 'Retail']
POL-002069 -> business_type : ['construction', 'Construction']
POL-002121 -> business_type : ['trucking', 'Trucking']
POL-002448 -> business_type : ['Retail', 'retail']
POL-002656 -> business_type : ['Trucking', 'TRUCKING']
POL-002720 -> business_type : ['Retail', 'RETAIL']
POL-003146 -> business_type : ['Service', 'service']
POL-003284 -> business_type : ['Construction', 'CONSTRUCTION']
POL-003664 -> business_type : ['trucking', 'Trucking']
POL-003847 -> business_type : ['Retail', 'retail']
POL-004008 -> business_type : ['Construction', 'construction']
POL-005401 -> business_type : ['retail', 'Retail']
POL-006638 -> business_type : ['Service', 'service']
POL-007227 -> business_type : ['Trucking', 'trucking']
POL-007695 -> business_type : ['Retail', 'retail']
```

business_type	
business_type	
Trucking	3367
Retail	2799
Construction	2285
Service	2278
trucking	696
retail	596
Other	573
TRUCKING	473
construction	435
service	431
RETAIL	387
CONSTRUCTION	286
SERVICE	286
other	100
OTHER	75
Agriculture	8



EDA - Data Quality

- checked numerical categories whether they contained negative numbers
- 5 records contained negative annual premiums
 - set to NaN

	policy_id	annual_premium	coverage_type	snapshot_date
2571	POL-003964	-8990.33	APD	2021-02-18
5362	POL-003525	-6266.99	APD	2020-11-14
10139	POL-005629	-6740.82	APD	2021-05-01
10327	POL-008752	-5780.46	AL	2021-10-24
14455	POL-013297	-11553.09	AL	2021-08-28



EDA — Claims & Coverage

12.36% overall claim rate

APD has higher claim frequency, while AL has substantially higher claim severity

	AL	APD
Policies	7,500	7,500
Claim rate	9.1%	15.6%
Avg. loss	\$18.2K	\$1.2K



EDA — Claims & Coverage

Claim frequency and loss severity vary across business types, making `business_type` a relevant candidate feature

Agriculture excluded from comparison due to very small sample size (n=8)

Business type	Claim rate	Avg. loss
Trucking	13.4%	\$13.7K
Construction	12.9%	\$9.8K
Retail	11.7%	\$8.3K
Service	11.4%	\$5.8K



EDA — Main Modeling Implications

APD and AL have different risk profiles

Claims are relatively sparse and losses are highly skewed

External risk score contains useful signal

Post-claim variables create leakage and must be excluded



Modeling methodology improvements

Develop separate APD and AL models to capture materially different frequency and severity patterns

First predict the likelihood of a claim, then predict the size of the loss if a claim happens

Use validation set to set the parameters of the model



Production readiness improvements

Make sure the model only uses information that is available when the policy starts, and prevent data leakage

Set up monitoring to track data quality, changes in the portfolio, predictions, and model performance over time

Build a reliable production pipeline with automated validation and scoring, model versioning, prediction logging, and clear rules for when the model should be reviewed or retrained



Deployment roadmap

Finalize the data and feature pipeline

- Clean and validate incoming data
- Make sure all features are available at policy inception
- Standardize preprocessing between training and production



Deployment roadmap

Validate the final model

- Compare APD and AL performance
- Test the model on the 2023–2024 scoring population



Deployment roadmap

Prepare for production

- Automate data validation, feature engineering and scoring
- Add model/version tracking and prediction logging
- Test the pipeline with realistic production data



Deployment roadmap

Set up monitoring

- Monitor data quality
- Monitor prediction distributions
- Once claims become available, monitor actual model performance



Deployment roadmap

Go-live and ongoing review

- Complete documentation and model governance approval
- Monitor the first production results
- Define clear triggers for model review, recalibration or retraining



AI tools

Tool used: ChatGPT

Used to support:

- **EDA:** suggesting data-quality checks and exploratory analyses
- **Methodology:** discussing modeling and validation approaches

All analysis and final decisions were manually reviewed and verified

Thank you!